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S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
ISE-II Examination Sem-II, AY 2025-26
Class: Final Year B.Tech (Instrumentation Engineering)

Date: 23/03/2026
Time: 14.30 to
15.15 pm

Subject: Elective: Intelligent Sensors
Code: PCC-IN432

Max. Marks: 20

Sr. No.	Questions	Marks	COs	BT
Q.1	With neat sketch explain the tactile sensor and its application.	05	CO5, CO2	BT1, BT2
Q.2	Draw the block diagram of components of machine vision system and explain each block in detail.	05	CO2, CO6	BT1, BT4
Q.3	How is optical fibre used for stress sensing? Describe a micro bend sensor and discuss its operation.	05	CO2	BT1, BT2
Q.4	Explain with neat diagram flow measurement using Fiber optic sensor.	05	CO2	BT1, BT2



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

Department of Instrumental Engineering

In Semester Examination-II

Academic Year: 2025-26 (semester-II)

Class: Fina Year B. Tech.	Subject: Chemical & Analytical Instrumentation	Subject Code: PCC-IN402
Date: 28/03/2026	Time: Time: 11.00 am – 11.45 am	Max. Marks: 20

Note:

- Assume suitable data if necessary and state clearly
- Mobile phones and programmable calculators are strictly prohibited
- Exchange/sharing of stationary, calculator etc. not allowed
- The symbols carry usual meaning.

Q		Marks	CO	BT
Q.1	All questions are compulsory			
a	Give the principle and schematic representation of gas chromatography.	3	CO1	BT2
b	How can alkali and alkaline earth metals be identified based on their flame colors and emitted wavelengths in a flame photometer?	3	CO1	BT3
c	Write short note on adsorption and thin-layer chromatography.	3	CO5	BT2
d	What is the difference between packed columns and capillary columns in GC?	3	CO2	BT1
Q.2	Solve any two questions			
a	Design a schematic representation of a gas chromatographic system and assess its applications across various scientific and industrial fields.	4	CO3	BT6
b	With the help of a neat, labelled diagram, evaluate the process of paper chromatography and determine how the different components of a mixture are separated based on their relative movement on the chromatographic paper.	4	CO4	BT5
c	Evaluate the strengths and weaknesses of the flame photometer in quantitative and qualitative metal analysis.	4	CO2	BT4

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

Course Outcomes



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri,
Nanded-431606.

Department of Instrumentation
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: Final Year B. Tech	Course Name: Modern Control Theory	Course Code: PCC-IN-401
Date: 30/03/2026	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- All Questions are compulsory

Q.N.		Marks	CO	BT
Q.1	$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ Find out state transition matrix using give state matrix A	10	CO4	BT4
Q.2	Show that the following system is not completely observable $\dot{x} = Ax + Bu, \quad y = Cx$ $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [4 \quad 5 \quad 1]$	10	CO4	BT4

=== ALL THE BEST ===



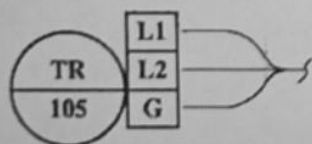
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Instrumentation Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – II)

Class: Final Year B.Tech	Subject: Instrumentation Project Management	Subject Code: PCC-IN421
Date: 25.03.2026	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.
- Solve any Two

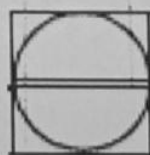
Q.N.		Marks	CO	BT
Q.1	a	Elaborate the purpose of P&ID diagram. Explain how P&ID is developed.	05	CO3 BT2
	b	With an example explain material balance sheet, also write how instrumentation engineer can use this document.	05	CO3 BT2
Q.2	a	Write a short note on installation sketch of pressure gauge.	05	CO4 BT1
	b	Elaborate what is GA drawing? Differentiate between plant lay out and GA drawing.	05	CO4 BT2
Q.3	a	What is a Loop Wiring Diagram? Explain it in detail.	05	CO4 BT1
	b	Identify the symbols in Figure A.	05	CO3 BT1



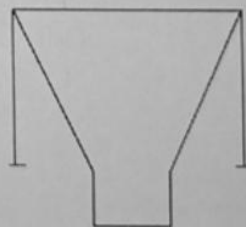
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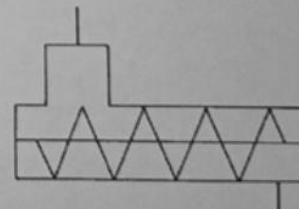
(ii)



(iii)



(iv)



(v)

Figure A